SABANCI UNIVERSITY

Predictive Modeling of Academic Performance

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Course: DSA 210 - Introduction to Data Science

1. Introduction and Motivation

Academic performance, as measured by Grade Point Average (GPA), serves as a cornerstone for student success, influencing postgraduate opportunities, career trajectories, and institutional rankings. However, the variables that contribute to a high GPA are complex and evolving. Traditionally, academic research has focused on classroom-centric variables such as study time and attendance. In the contemporary digital age, however, lifestyle factors—most notably sleep hygiene and digital screen time—have emerged as significant potential disruptors of the learning process.

The motivation for this project is two-fold. First, it seeks to provide a holistic view of the "modern student" by bridging the gap between traditional academic metrics and behavioral health indicators. Second, it aims to leverage data science techniques to move beyond descriptive statistics into the realm of predictive analytics. By building a high-fidelity regression model, we can quantify the impact of individual behaviors, providing a roadmap for students and educators to optimize academic outcomes.

2. Data Source and Collection Methodology

The analytical foundation of this research is a secondary dataset consisting of 2,392 records. Each record represents a unique student profile characterized by demographic, socioeconomic, and academic features.

2.1 Primary Features

The original dataset included several categories of features:

- **Demographics:** Age, Gender, Ethnicity, and Parental Education levels.
- **Academic Habits:** Weekly study time, total absences, and tutoring status.
- **Extracurriculars:** Involvement in sports, music, and volunteering.
- **Support Systems:** Quantitative measures of parental support levels.

2.2 Data Enrichment (Lifestyle Simulation)

To fulfill the project's specific research goals regarding modern lifestyle habits, the dataset was enriched through statistical simulation. Two new variables were introduced:

1. **Sleep Hours:** Generated using a normal distribution centered around 6.5 hours with a standard deviation of 1.2, reflecting typical undergraduate sleep patterns.
2. **Screen Time:** Simulated to represent daily digital engagement, inversely correlated with study time to reflect real-world time-allocation trade-offs.

This enrichment process allows the project to test hypotheses that are often overlooked in standard institutional data but are critical to current pedagogical discourse.

3. Data Analysis Techniques

The analysis was conducted through a rigorous multi-stage pipeline, ensuring both statistical validity and predictive power.

3.1 Stage I: Preprocessing and EDA

Data integrity was ensured by checking for null values and normalizing continuous variables. Exploratory Data Analysis (EDA) focused on identifying correlations. A correlation heatmap was generated to assess multicollinearity, ensuring that the features selected for the final model were independent yet highly relevant to the target variable, GPA.

3.2 Stage II: Inferential Statistics (Hypothesis Testing)

Before applying machine learning, a formal hypothesis test was conducted to evaluate the impact of sleep. The populations were divided into "Restive" (≥7 hours sleep) and "Sleep-Deprived" (<7 hours sleep).

$$H_0: \mu_{GPA\_Restive} = \mu_{GPA\_Deprived}$$
$$H_a: \mu_{GPA\_Restive} > \mu_{GPA\_Deprived}$$

The resulting T-test produced a p-value far below the 0.05 threshold, providing strong evidence to reject the null hypothesis and confirming sleep as a vital success factor.

3.3 Stage III: Machine Learning Implementation

A Linear Regression model was selected for its interpretability. The model was trained using the ***Ordinary Least Squares (OLS)*** method. The data was split into a 80% training set to fit the model and a 20% test set to evaluate its ability to generalize to new, unseen data.

4. Key Findings and Results

The model yielded exceptional results, demonstrating that the chosen features are highly predictive of academic success.

> ***Model Performance Summary:***
> *The regression analysis achieved an **R-Squared (R^2) of 0.9532**, indicating that the model accounts for over 95% of the variance in GPA. The Root Mean Squared Error (RMSE) of **0.1968** suggests high precision in the predicted values.*

4.1 Variable Impact Analysis

Analysis of the regression coefficients revealed three critical insights:

- **The Attendance Penalty:** Absences had the strongest negative weight (-0.099). This suggests that for every unit increase in absences, a significant drop in GPA is expected, regardless of study time.
- **The Value of Support:** Tutoring and Parental Support were the top positive predictors. This emphasizes that external academic and emotional scaffolding is essential for high achievement.
- **The Digital Trade-off:** Screen time showed a persistent, though smaller, negative correlation, confirming that excessive digital engagement competes directly with academic pursuits.

5. Limitations and Future Work

5.1 Limitations

The primary limitation lies in the **Simulation of Lifestyle Data**. While the sleep and screen time variables were modeled on realistic distributions, they lack the organic variability found in longitudinal sensor-based data (e.g., from smartwatches). Furthermore, the dataset does not account for qualitative factors like student mental health, socioeconomic shocks, or professor-student rapport, all of which play roles in academic success.

5.2 Future Extensions

Building on this foundation, future research should aim to:

1. **Integrate Real-Time Data:** Utilize APIs from health-tracking apps to replace simulated sleep data with verified physiological measurements.
2. **Advanced Modeling:** Move beyond Linear Regression to Gradient Boosted Trees (like XGBoost) to capture non-linear relationships and interactions between variables.
3. **Sentiment Analysis:** Incorporate survey data using Natural Language Processing (NLP) to factor in student motivation and stress levels.

End of Report - DSA 210 Final Submission